# Connect the car

Read the named terminals here; page 2 locates the actual solder pads. All grounds connect together. Crossings connect only where marked with a dot.

## 1 BATTERY


2S / 7.4 V / 300 mAh

8.4 V when fully charged

+

-

XT30

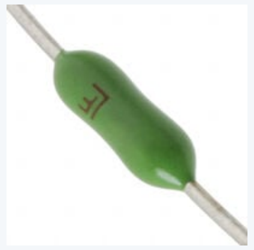
unplug = OFF

FUSE

4 A*

IN+

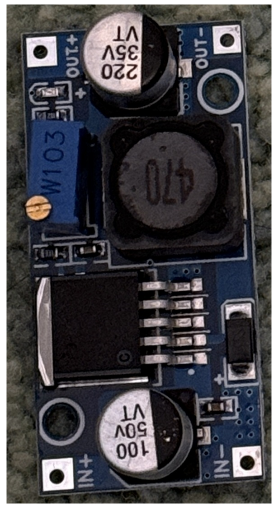
IN-



Fuse family photo

Details: page 6

## 2 YOUR LARGE BUCK



OUT+ to OUT- = 5.00 V

OUT+ / 5 V BUS

OUT- / GND BUS

GPIO16 > LEFT SIGNAL

GPIO18 > RIGHT

## 3 FRONT-LEFT SERVO

RED +

BROWN -

S


Owned continuous-rotation unit

## 4 REAR-RIGHT SERVO

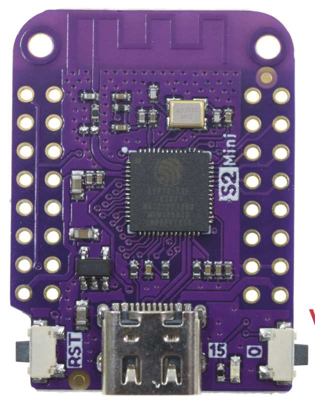
RED +

BROWN -

S


Same supply; separate signal

## 5 LOLIN S2 MINI




VBUS

J_PWR

LINK

GND

5 V enters VBUS, never 3V3

## At the 5 V / GND split

At the split: **220 uF / 10 V electrolytic** (+ to 5 V; striped negative to GND). At the S2, **after J_PWR:** add **100 nF ceramic across VBUS and GND** with short leads. Preserve the buck's onboard capacitor.

## Battery sensing: included on page 4

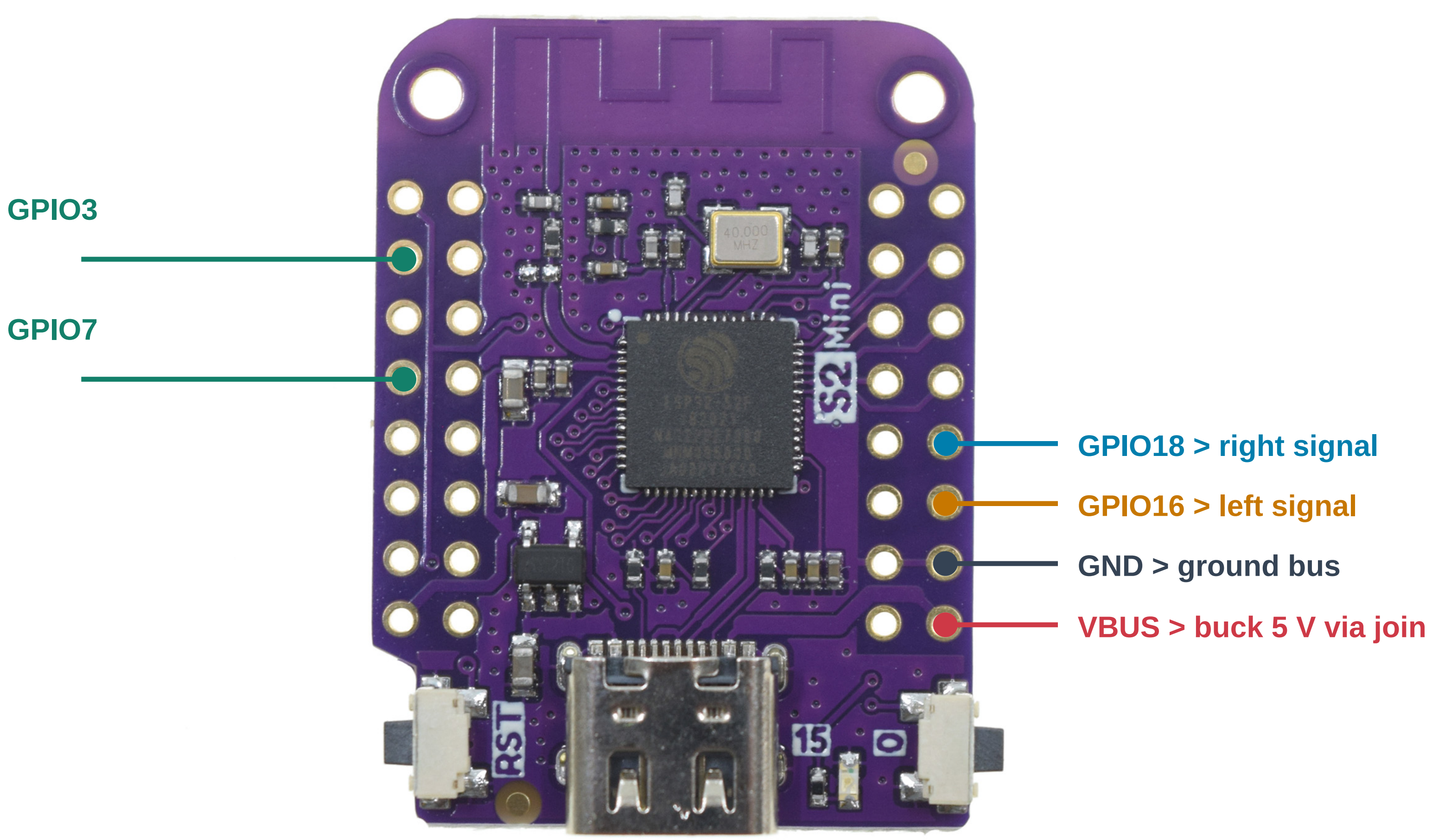
The existing receiver firmware also needs **GPIO3 + GPIO7** and the switched divider. It stops motion at 7.0 V; it does **not** disconnect the pack. Unplug after stopping.

* Fuse rating is provisional. Photos show reference servos; wire functions, not plug orientation, define the connections.

# Exactly where each wire lands

Component side facing you. S2 antenna at the top; USB-C at the bottom. Photos are not mirrored.

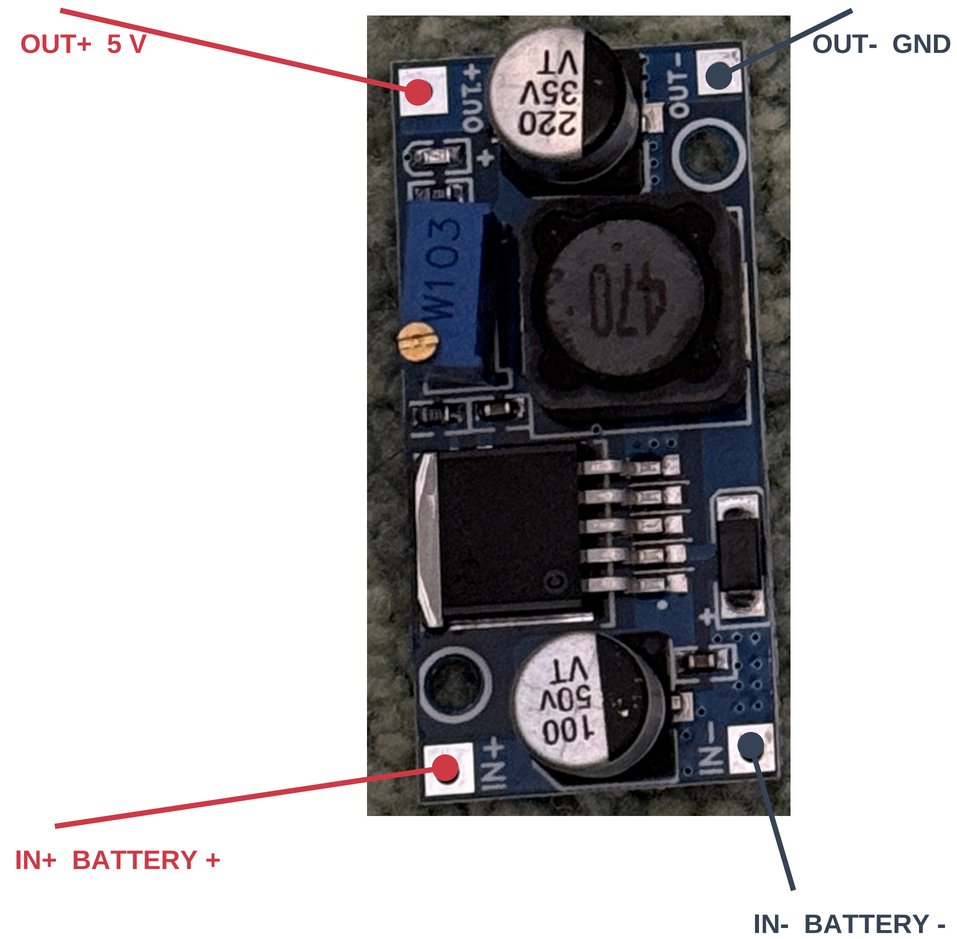
### LOLIN S2 MINI v1.0.0 REFERENCE




Left GPIO3 / GPIO7: sensing only (page 4)

**Use GPIO labels. Inner-row GPIO17 is NOT the right servo pin.**

### YOUR LARGE BUCK: SAME ORIENTATION




Blue trimmer sets output. Measure OUT+ to OUT- and set **5.00 V with all loads disconnected**. IC marking is not readable: confirm the board type.

## Soldered wiring + removable connectors

Bare S2 holes and converter pads need soldered wires or soldered headers. Use 22 AWG or thicker for the battery and main 5 V / GND paths. Servo red normally means supply; brown/black ground; orange/yellow signal. Verify your servo wires before fitting a 3-pin extension. Split its three wires to the correct nets - do not plug the whole servo lead onto random adjacent S2 pins.

# Battery, plug and converter choice

Reuse the larger buck for this prototype. A custom power PCB is not required.


## BUY: LUMENIER 300

**2S 75C LiPo, XT30**
48 x 17 x 12 mm
18 g; 7.4 V nominal
8.4 V fully charged

$13.49; listed available
at the research snapshot.

RaceDayQuads product / purchase page

## YOUR YELLOW CONNECTORS

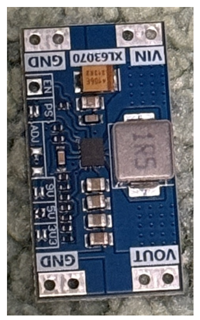
The photo establishes the XT family, but does not distinguish **XT30 from XT60**. Read the molded marking.

**XT30:** use the mating harness half.
**XT60:** buy an XT30 mating pigtail; a bulky adapter also works but increases footprint.

Check + / - with a meter. Housing shape or wire color alone is not proof of polarity.

## Why the large buck wins here

A 2S pack feeds a step-down converter naturally. Allowing 1.2 A per servo plus 0.4 A for the S2 gives a **provisional 2.8 A at 5 V**. Actual servo peaks are unknown. The LM2596 IC is rated 3 A with suitable parts and cooling; this particular board must still pass load and temperature checks at 8.4 V and 7.0 V.



## Why not the small XL63070 board?

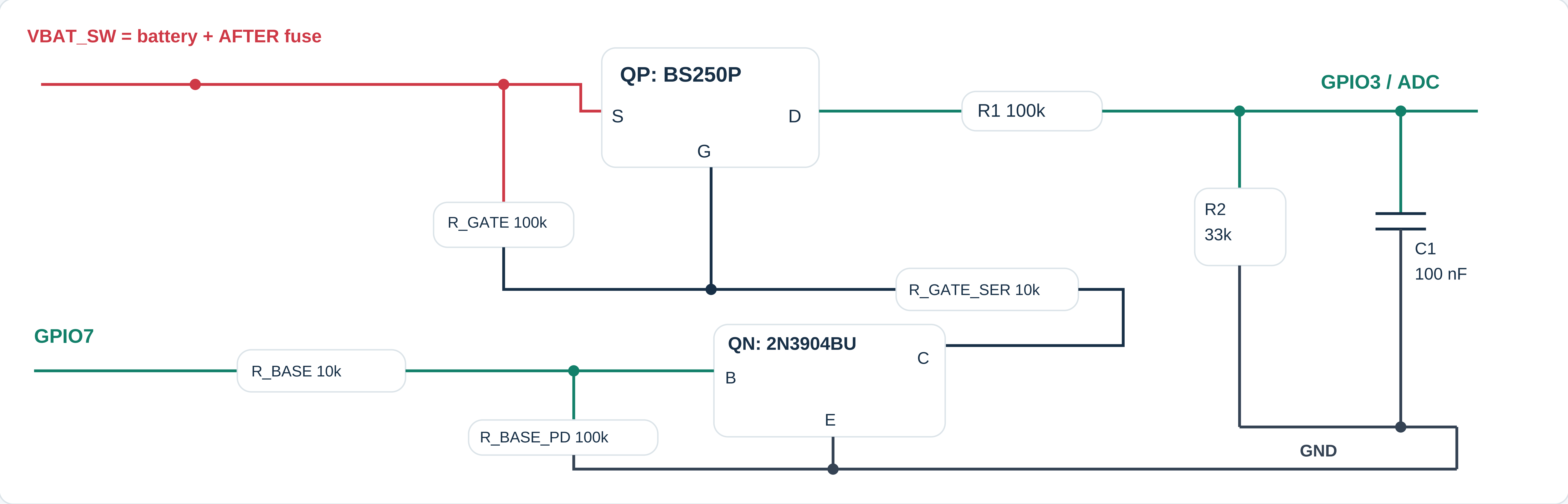
It appears to be a TPS63070-style buck-boost module; the IC is unconfirmed. TI specifies 2 A output at 4 V in / 5 V out. That is below our provisional load. A 1S design at 14 W could also need about 5 A near 3.3 V. Smaller hardware needs current measurements first.

## Battery handling and fit

12 mm is the published nominal pack height, not measured installed height: leave room for padding and lead bends. This hobby pack has no verified protection PCB. Neither owned converter provides charging, balancing or a suitable LiPo cutoff. Use a 2S balance charger and unplug the pack after each run.

# Battery sensing for the existing firmware

A small insulated perfboard carries these low-current parts; motor current stays in the main harness.




## Parts and terminal identification

**QP:** Diodes BS250P P-channel MOSFET: source to raw fused battery, drain to R1, gate to the resistor network shown.
**QN:** onsemi 2N3904BU NPN: emitter to GND, collector to R_GATE_SER, base to R_BASE.
**Resistors:** 3 x 100k, 2 x 10k, and 1 x 33k; use 1% for R1/R2. C1: 100 nF ceramic, 16 V or higher.
Match the exact maker's package-view drawing before soldering; transistor terminal letters are not left-to-right lead positions.

## What it does - and what it cannot do

GPIO7 HIGH enables sensing; wait at least 20 ms before sampling GPIO3. Return GPIO7 LOW afterward.
**8.4 V pack = about 2.084 V at GPIO3.**
The switching stage avoids a sustained battery-fed ADC path when the S2 is off; do not replace it with a permanently connected divider.
Browser-drive firmware needs servo-neutral and battery calibration before arming.
Pairing is only for the optional handheld. Its 7.0 V stop only commands neutral. Pack voltage does not reveal an individually weak cell; monitor both cells and unplug promptly.

# Build, check and use

No custom PCB. This is a soldered prototype harness, not a tested plug-and-play electrical assembly.

## Wire and check in this order

1. Battery unplugged: solder the mating XT30 lead through a provisional 4 A inline fuse to IN+; negative to IN-. Insulate every joint.

2. Leave all loads disconnected. Power the buck, read OUT+ to OUT- with a multimeter and adjust to 5.00 V. Unplug before adding wires.

3. Split OUT+ and OUT- into separate servo and S2 branches. Add the capacitors. Fit J_PWR as a removable insulated connector in the S2 VBUS feed.

4. Add the page 4 sensing circuit. Check the ADC voltage and off-state behavior before connecting GPIO3. Calibrate against a meter.

5. With wheels lifted, test each servo, then both together. Check 5 V stability, resets and board temperature at 8.4 V and 7.0 V input. Do not hold a servo stalled.

6. Verify neutral on signal loss and the calibrated 7.0 V stop. Test voltage thresholds with an adjustable supply, not by deeply discharging the battery.

## Before connecting USB to the S2

**Unplug the battery, remove J_PWR, and disconnect both servo signal wires.** Ground may remain connected. The S2 VBUS header is connected to USB power: battery disconnection alone can still let USB feed the servos or buck backward through the 5 V wire. Restore J_PWR and signals only after USB is removed.

## Charge outside the car

Use a **2S LiPo balance charger** set for 4.20 V/cell (8.40 V pack). A conservative 1C setting for 300 mAh is **0.30 A**, subject to pack instructions. Connect its XT30 main lead and 3-pin balance lead as the charger requires. A 1S USB/TP4056 charger is not compatible. Charge attended on a nonflammable surface.

**POC battery checks:** check both cells with your multimeter before and between short, attended test runs. Do not bridge adjacent balance contacts with probe tips. Stop and unplug when firmware stops; do not keep restarting to drain the pack further. Manual checks and pack-voltage sensing do not provide automatic power cutoff.

## Sources, image credits and build files

WEMOS: S2 Mini photo, board pinout and schematic

TI: LM2596 buck datasheet

TI: TPS63070 buck-boost datasheet

TowerPro: MG90S reference image (not proof of continuous rotation)

Diodes: BS250P terminal drawing

onsemi: 2N3904BU terminal drawing

# What is underneath the deck?

The fuse protects the input wiring; the small perfboard measures battery voltage. Neither is a charger.

## F_IN: a compact fuse candidate

251-series reference photo
Appearance / markings may vary

**Littelfuse 0251004.MXL**
PICO II 251 series
**4 A, very fast acting**
125 V DC rated
300 A DC interrupt rating

Body: 7.11 mm long,
2.80 mm maximum diameter.
Leads and insulation add space.

DigiKey: 0251004.MXL / purchase page

Littelfuse: specifications and mechanical drawing

## Why these extra parts are here

**Fuse:** opens the battery-positive feed during a sufficiently large overcurrent, such as a wiring short. It is one-time use; replace after finding the fault. It is not low-battery protection or a precise 4 A current limiter.

**Perfboard:** a small soldering board holding the page 4 voltage divider and two-transistor enable circuit. GPIO3 reads scaled battery voltage; GPIO7 enables the reading. This supports the existing firmware monitoring. It carries no servo current.

**Capacitors:** the nearby 220 uF and 100 nF parts help smooth the 5 V supply. They do not replace an adequately sized converter.

## Where the fuse connects

XT30 harness + — F_IN fuse — VBAT_SW: buck IN+ and page 4 sensing feed

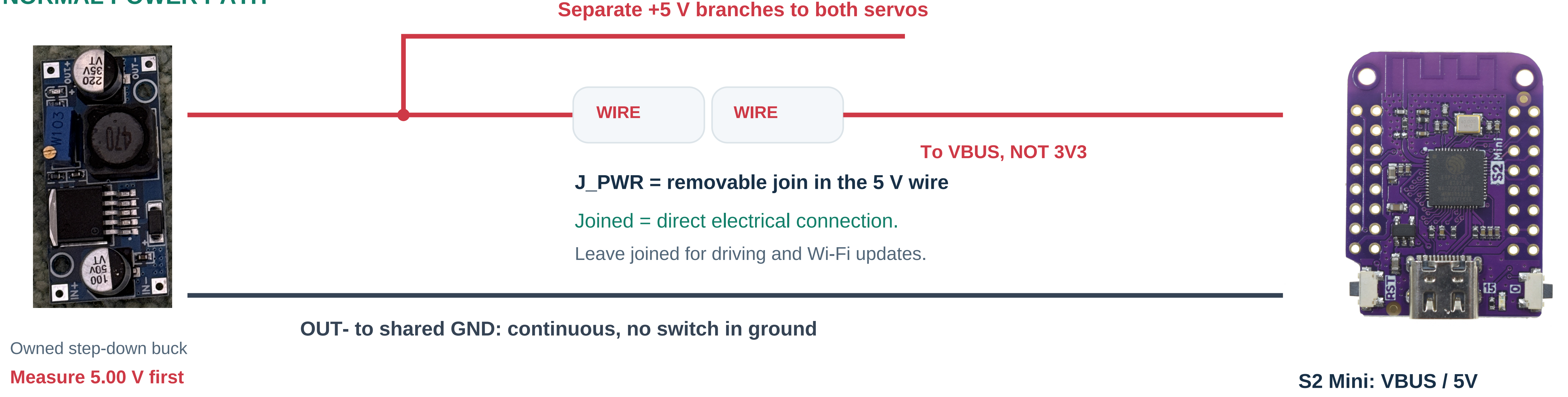
**Assembly:** put the fuse close to the battery connector in the positive harness lead. This axial candidate is soldered inline and insulated with strain relief; it does not require a bulky cartridge holder. Either fuse lead can face the battery. Keep motor current out of the sensing perfboard.

**Still provisional:** validate 4 A against measured startup current, wire size and fault current. The updated Blender body follows the axial candidate; lead bends, insulation and loose cradle retention still need a fit check.
Photo: Littelfuse / DigiKey, reference image supplied with this product listing.

# Buck 5 V goes straight to ESP32 VBUS.

J_PWR is our label for a removable join in that wire. It is not a board, regulator, or ESP32 pin.

## NORMAL POWER PATH




### Driving on your home Wi-Fi

Set your network in **web.private.h**. Open **http://spy-car.local/** on a computer or phone on the same network. Your computer keeps internet access. Battery, buck, J_PWR and servos stay connected; no USB cable.

### Wireless update mode: current firmware

Lift the wheels. Boot normally, then hold **BOOT/0 for 3 seconds**. Driving locks off. Join **SpyCar-Update**; open **http://192.168.4.1/update**. Login: **admin**. Use your configured password for both Wi-Fi and login. Keep USB unplugged.

### USB recovery and the next software step

**USB initial setup / recovery:** unplug XT30, separate J_PWR and unplug both servo signals first. Ground may remain connected. **Planned, not yet implemented:** updates over home Wi-Fi at spy-car.local/update, with SpyCar-Update as fallback. The wiring already supports both approaches; no extra switch is needed.